

HW02 - Received Power and Link Geometry

ENAE694: Spacecraft Communications

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Problem 1**Slant Range (20 pts.)**

A spacecraft is in a circular orbit 500 km above Earth. Calculate its slant range from a ground station when its elevation angle is 10° . Use an Earth radius of 6378 km, neglect station altitude, and round to the nearest kilometer. Set your calculator to degrees.

The slant range at elevation e is

$$d = \sqrt{(R_E + h)^2 - R_E^2 \cos^2(e)} - R_E \sin(e)$$

Substituting $R_E = 6378$ km, $h = 500$ km, and $e = 10^\circ$ gives

$$\begin{aligned} d &= \sqrt{(6378 \text{ km} + 500 \text{ km})^2 - (6378 \text{ km})^2 \cos^2(10^\circ)} \\ &\quad - (6378 \text{ km}) \sin(10^\circ) \\ &= 1695.081 \text{ km} \\ &\approx 1695 \text{ km} \quad \blacksquare \end{aligned}$$

Thus, the spacecraft's slant range is 1695 km to the nearest kilometer.

Problem 2**Voyager 2 Received Power (60 pts.)**

Calculate total received signal power for Voyager 2's X-band link to DSS-43, the 70 m antenna in Canberra. Use the updated range below with radio parameters from JPL's January 1, 1996 budget [1, Table 5-3, p. 26].

- Range: 21 476 670 660 km
- Frequency: 8.415 GHz
- Transmit Power: 12.3 W
- Transmit Antenna Gain: 48.2 dBi
- Receive Antenna Gain: 74.01 dBi

Use 0 dB transmit feed loss and 0.42 dB other losses (pointing, atmosphere, and polarization combined).

- Calculate [EIRP] in dBW. (20 pts.)
- Calculate free-space path loss [L_p] in dB. (20 pts.)
- Calculate received power [P_r] in dBW. Round your results to one decimal place. (20 pts.)

First, the transmitter power in logarithmic units is

$$[P_t] = 10 \log_{10} \left(\frac{12.3 \text{ W}}{1 \text{ W}} \right) = 10.899 \text{ dBW}$$

a) EIRP:

With zero transmit feed loss,

$$\begin{aligned} [EIRP] &= [P_t] - [L_t] + [G_t] \\ &= 10.899 \text{ dBW} - 0 \text{ dB} + 48.2 \text{ dBi} \\ &= 59.1 \text{ dBW} \quad \blacksquare \end{aligned}$$

b) Free-space path loss:

$$\begin{aligned} [L_p] &= 92.45 \text{ dB} + 20 \log_{10}(21476670660) + 20 \log_{10}(8.415) \\ &= 317.6 \text{ dB} \quad \blacksquare \end{aligned}$$

c) Received power:

Subtracting the combined pointing, atmospheric, and polarization loss,

$$\begin{aligned} [P_r] &= [EIRP] - [L_p] + [G_r] - [L_{\text{other}}] \\ &= 59.1 \text{ dBW} - 317.6 \text{ dB} + 74.01 \text{ dBi} - 0.42 \text{ dB} \\ &= -184.9 \text{ dBW} \quad \blacksquare \end{aligned}$$

Therefore, the total received signal power is -184.9 dBW.

Problem 3**The Elevation Mask Trade (20 pts.)**

For the Earth-orbiting spacecraft in Problem 1, suppose we increase the minimum elevation angle used for contacts. In two or three sentences, explain what happens to the maximum useful slant range, the worst-case free-space path loss, and the available contact time. Keep the orbit, station, frequency, and antenna gains unchanged.

Increasing the minimum elevation angle decreases the maximum useful slant range, so the worst-case free-space path loss during an accepted contact also decreases. However, the spacecraft spends less of each pass above the higher elevation mask, so the available contact time decreases.

Code

```
./src/index.py Python
1  import math
2  from pathlib import Path
3
4  ROOT = Path(__file__).resolve().parent.parent
5  TEXT_DIR = ROOT / "outputs" / "text"
6
7
8  def heading(problem: str) -> None:
9      """Print a clearly delimited problem heading."""
10     print(f"\n-----\n {problem}\n-----")
11
12
13  def write_result(name: str, value: str) -> None:
14      """Write one value for direct inclusion in the Typst report."""
15      TEXT_DIR.mkdir(parents=True, exist_ok=True)
16      (TEXT_DIR / f"{name}.txt").write_text(f"{value}\n", encoding="utf-8")
17
18
19  def problem_1() -> None:
20      """Calculate slant range at a specified elevation angle."""
21      heading("p01")
22
23      earth_radius_km = 6_378.0
24      altitude_km = 500.0
25      elevation_deg = 10.0
26      elevation_rad = math.radians(elevation_deg)
27
28      slant_range_km = (
29          math.sqrt(
30              (earth_radius_km + altitude_km) ** 2
31              - earth_radius_km**2 * math.cos(elevation_rad) ** 2
32          )
33          - earth_radius_km * math.sin(elevation_rad)
34      )
35      rounded_range_km = round(slant_range_km)
36
37      print(f"Slant range (unrounded): {slant_range_km:.6f} km")
38      print(f"Slant range (nearest km): {rounded_range_km} km")
39      write_result("s01", str(rounded_range_km))
40
41
42  def problem_2() -> None:
43      """Calculate the Voyager 2 X-band link budget."""
44      heading("p02")
```

```
45
46     range_km = 21_476_670_660.0
47     frequency_ghz = 8.415
48     transmit_power_w = 12.3
49     transmit_gain_dbi = 48.20
50     receive_gain_dbi = 74.01
51     transmit_feed_loss_db = 0.0
52     other_losses_db = 0.42
53
54     transmit_power_dbw = 10.0 * math.log10(transmit_power_w)
55     eirp_dbw = transmit_power_dbw - transmit_feed_loss_db + transmit_gain_dbi
56     path_loss_db = (
57         92.45
58         + 20.0 * math.log10(range_km)
59         + 20.0 * math.log10(frequency_ghz)
60     )
61     received_power_dbw = (
62         eirp_dbw - path_loss_db + receive_gain_dbi - other_losses_db
63     )
64
65     print(f"Transmit power: {transmit_power_dbw:.6f} dBW")
66     print(f"EIRP: {eirp_dbw:.1f} dBW")
67     print(f"Free-space path loss: {path_loss_db:.1f} dB")
68     print(f"Received power: {received_power_dbw:.1f} dBW")
69     write_result("s02a", f"{eirp_dbw:.1f}")
70     write_result("s02b", f"{path_loss_db:.1f}")
71     write_result("s02c", f"{received_power_dbw:.1f}")
72
73
74 def problem_3() -> None:
75     """Problem 3 is qualitative and requires no numerical computation."""
76     heading("p03")
77     print("No numerical output: see the qualitative explanation in the report.")
78
79
80 def main() -> None:
81     problem_1()
82     problem_2()
83     problem_3()
84
85
86 if __name__ == "__main__":
87     main()
```

References

Formulas

$$d = \sqrt{(R_E + h)^2 - R_E^2 \cos^2(e)} - R_E \sin(e) \quad (1)$$

$$[P] = 10 \log_{10} \left(\frac{P}{1W} \right) \quad (2)$$

$$[EIRP] = [P_t] - [L_t] + [G_t] \quad (3)$$

$$[L_p] = 92.45 + 20 \log_{10}(R) + 20 \log_{10}(f) \quad (4)$$

$$[P_r] = [EIRP] - [L_p] + [G_r] - [L_{\text{other}}] \quad (5)$$

For slant range, h is altitude and e is elevation. In the path-loss formula, enter R in km and f in GHz. Brackets denote decibel values. Power is in dBW, antenna gains are in dBi, and losses are in dB. Record losses as positive amounts and subtract them.

Bibliography

- [1] R. Ludwig and J. Taylor, "Voyager Telecommunications," Jet Propulsion Laboratory, California Institute of Technology, Pasadena, California, DESCANSO Design and Performance Summary Series 4, Mar. 2002. [Online]. Available: https://descanso.jpl.nasa.gov/DPSummary/Descanso4-Voyager_new.pdf